

ReTR: Contrastive Vision-Language Alignment for Render-to-Material Texture Retrieval

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Abstract

Retrieving a physically-based rendering (PBR) material texture from a large library given only a rendered image of a 3D object is a challenging cross-modal matching problem: the query is a composite scene image that conflates geometry, lighting, and material appearance, while the gallery contains flat, uniformly-lit texture maps. We address this *render-to-material* retrieval task by contrastively fine-tuning the visual backbone of InternVL2-1B, a state-of-the-art vision-language model (VLM). A lightweight two-layer projection head maps the CLS token of the unfrozen Vision Transformer (ViT) into a shared 256-dimensional embedding space, trained with a symmetric InfoNCE loss and a learned temperature. We construct a paired dataset of **19,568** (render, texture) pairs and propose **RMR-Score**, a unified evaluation metric that combines precision, coverage, and ranking quality into a single scalar. On the full validation gallery of 1,956 textures, the fine-tuned model achieves $R@1 = 34.5\%$, $R@5 = 71.6\%$, $nDCG@10 = 0.579$, and $RMR\text{-Score} = 0.524$, outperforming CLIP ViT-L/14 by $19\times$ on $R@1$.

1 Introduction

Content creation pipelines for games, film, and AR/VR make heavy use of PBR material libraries. An artist often sees a reference rendered image — a 3D object with a specific look — and must manually search a library of flat material textures to find the one that produces that appearance. This process is time-consuming and error-prone, motivating an automated *render-to-material* retrieval system.

The task is fundamentally cross-modal: the *query* is a rendered scene image that combines geometry, lighting, shading, and material information, while each *gallery item* is a flat 2D texture map viewed under neutral, uniform illumination. Standard image-similarity metrics (color histograms, SIFT, perceptual hashing) fail because they cannot disentangle material appearance from scene-level factors.

Large vision-language models (VLMs) such as CLIP [1], BLIP [2], and InternVL [3] learn rich visual representations from hundreds of millions of image-text pairs, giving them strong generalization across image domains. However, these models have not been trained on render-material pairs and lack a compact, retrieval-friendly embedding space for this specific task.

Moreover, existing cross-modal retrieval benchmarks evaluate with *isolated* metrics — Recall@ K , MRR, or median rank — making it difficult to holistically compare methods that trade off precision against coverage. No unified metric exists for this setting.

In this paper we ask: *can a VLM visual backbone be efficiently adapted to align rendered scene images with their source material textures, and how should retrieval quality be measured?* We answer both questions by (i) fine-tuning only the ViT encoder and a small projection head of InternVL2-1B with a contrastive objective, and (ii) proposing **RMR-Score**, a single scalar that jointly captures precision ($R@1$), coverage ($R@K$), and ranking quality (MRR). Our contributions are:

1. The first **render-to-material retrieval benchmark**: a paired dataset of 19,568 samples spanning diverse material categories, an evaluation protocol, and standardised baselines.
2. **RMR-Score**: a unified evaluation metric for cross-modal material retrieval that combines $R@1$, $R@K$, and MRR into a single comparable scalar, addressing the fragmented metrics problem in existing retrieval benchmarks.
3. A **contrastive fine-tuning recipe** for VLM visual backbones that uses a learned-temperature InfoNCE loss, gradient accumulation, and ViT unfreezing — fitting on a single consumer GPU.
4. Comprehensive **evaluation** showing that our method achieves $RMR\text{-Score} = 0.524$ and $R@5 = 71.6\%$, a $19\times$ $R@1$ improvement over the strongest zero-shot baseline.

2 Related Work

Contrastive Representation Learning. Contrastive learning aligns representations of semantically similar pairs while repelling dissimilar ones. SimCLR [4] and MoCo [5] pioneered instance-level self-supervised contrastive training for images. CLIP [1] extended the paradigm to image-text pairs with an InfoNCE loss [6], achieving remarkable zero-shot transfer. Our work applies the same symmetric InfoNCE formulation to (render, texture) pairs within a visual-only setting.

Vision-Language Model Backbones. InternVL [3] is a family of VLMs that pair a large InternViT encoder (up to 6B parameters) with an instruction-tuned large language model via a two-layer MLP projector. InternVL2-1B uses a 300M-parameter ViT with hidden size $d = 1024$ and achieves competitive performance on multimodal benchmarks. Because our retrieval task is purely visual, we discard the language model after fine-tuning, achieving a compact ViT-only inference pipeline.

Image-Based Material Retrieval. Traditional approaches rely on handcrafted features or CNN-based classifiers [7]. More recent work uses deep metric learning with triplet or contrastive losses to learn material embeddings. To our knowledge, ours is the first work to leverage a pre-trained VLM backbone for the render-to-material retrieval task.

Retrieval Evaluation Metrics. Cross-modal retrieval is typically evaluated with $\text{Recall}@K$, MRR, and median rank [1]. Some works report $rSum = R@1 + R@5 + R@10$ as a naive aggregate, but this ignores ranking quality and treats all K values equally. Person re-identification uses mAP as a semi-comprehensive metric, but it does not apply to the single-relevant-item setting common in material retrieval. We propose RMR-Score to address this gap.

3 RMR-Score: A Unified Retrieval Metric

Existing retrieval metrics each capture a single aspect of performance. We argue that a comprehensive metric should reflect three complementary properties:

1. **Precision:** the ability to place the correct answer at rank 1 (captured by $R@1$).
2. **Coverage:** whether the correct answer appears within a practical retrieval window (captured by $R@K$, we use $K=5$).

3. **Ranking quality:** how close to rank 1 the correct answer appears on average (captured by MRR).

We define the **Render-Material Retrieval Score** (RMR-Score) as:

$$\text{RMR} = \frac{w_1 \cdot R@1 + w_2 \cdot R@K + w_3 \cdot \text{MRR}}{w_1 + w_2 + w_3} \quad (1)$$

where $R@1$, $R@K$, and MRR are all in $[0, 1]$ and $w_1, w_2, w_3 > 0$ are importance weights. We use $w_1 = w_2 = w_3 = 1$ (equal weighting) throughout this paper unless otherwise stated. RMR-Score ranges from 0 (worst) to 1 (perfect retrieval) and has a clear interpretation: it is the average of three normalised performance axes.

Properties. (i) *Sensitivity:* RMR responds to changes in all three axes — a method that achieves high $R@5$ but poor $R@1$ (i.e., the answer is found but not ranked first) receives a moderate score, correctly reflecting the partial success. (ii) *Discriminability:* unlike rSum, RMR normalises to $[0, 1]$ and weighs the three aspects explicitly, preventing $R@10$ from dominating. (iii) *Simplicity:* the metric is easy to compute and report alongside standard metrics.

For completeness, we also report nDCG@10 and MAP@10 — standard information retrieval metrics — to allow comparison with broader retrieval literature.

4 Dataset

4.1 Data Collection

Each sample in our dataset is a pair (f, t) :

- **f (fig):** a rendered image of a 3D object with a specific material applied, captured under standard lighting conditions. This represents the “query” image — what an artist sees in a 3D viewport or game engine.
- **t (texture):** the corresponding flat PBR material map (albedo / diffuse channel), displayed under uniform neutral illumination. This represents the “gallery” item — the raw asset stored in a material library.

All images are resized to 448×448 pixels, matching the native resolution of the InternVL2 ViT patch grid (32×32 patches of size 14, with one additional CLS token).

4.2 Statistics

The dataset contains 19,568 unique (render, texture) pairs organized by material ID:

Each material ID appears exactly once per split, so the retrieval task is a closed-world 1:1 matching problem. The validation gallery therefore contains 1,956 candidate textures.

Table 1: Dataset splits.

Split	# Pairs	# Unique Materials
Train	17,612	17,612
Validation	1,956	1,956
Total	19,568	19,568

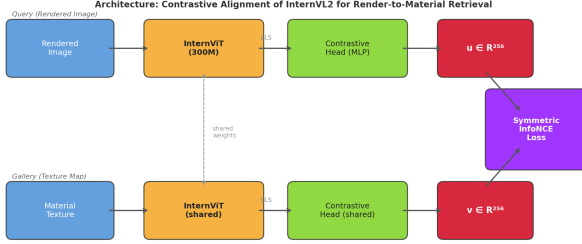


Figure 1: Architecture overview. Both the rendered image (query) and the texture map (gallery item) are processed by the same ViT backbone followed by the ContrastiveHead. Representations are aligned with a symmetric InfoNCE loss.

5 Method

5.1 Model Architecture

Figure 1 provides an overview of the system. We build on the visual backbone of **InternVL2-1B**, comprising a ViT with hidden size $d_{\text{ViT}} = 1024$ and 448×448 input resolution. The language model and MLP projector are discarded.

On top of the ViT we attach a **ContrastiveHead**:

$$h(\mathbf{x}) = \ell_2(\mathbf{W}_2 \sigma(\mathbf{W}_1 \text{CLS}(\mathbf{x}))), \quad (2)$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_{\text{ViT}} \times d_{\text{ViT}}}$, $\mathbf{W}_2 \in \mathbb{R}^{d_{\text{ViT}} \times 256}$, σ is ReLU, and ℓ_2 denotes L2 normalisation. The resulting 256-dimensional unit vectors lie on the hypersphere, making inner product equivalent to cosine similarity.

5.2 Contrastive Training Objective

Given a mini-batch of B (render, texture) pairs $\{(\mathbf{f}_i, \mathbf{t}_i)\}_{i=1}^B$, we compute embeddings:

$$\mathbf{u}_i = h(\mathbf{f}_i), \quad \mathbf{v}_i = h(\mathbf{t}_i). \quad (3)$$

The symmetric InfoNCE (NT-Xent) loss is:

$$\mathcal{L} = \frac{1}{2} [\ell(\mathbf{U}, \mathbf{V}) + \ell(\mathbf{V}, \mathbf{U})], \quad (4)$$

where

$$\ell(\mathbf{A}, \mathbf{B}) = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\mathbf{a}_i^\top \mathbf{b}_i / \tau)}{\sum_{j=1}^B \exp(\mathbf{a}_i^\top \mathbf{b}_j / \tau)}. \quad (5)$$

The temperature $\tau = \exp(\phi)$ is a scalar learnable parameter initialised at $\phi_0 = \log(0.07) \approx -2.66$ and clamped to $[e^{-4.6}, e^{2.3}]$ after each optimiser step to prevent collapse. To stabilise early training, ϕ is frozen for the first two epochs and then jointly optimised with the rest of the parameters.

5.3 Training Setup

The full training procedure is described in Algorithm 1. We use the **AdamW** optimiser [8] with weight decay 10^{-2} and a cosine learning rate schedule. Gradient clipping with max-norm 1.0 is applied at every optimiser step.

Algorithm 1 Contrastive Fine-Tuning of InternVL

Require: Pre-trained ViT f_θ , learning rate η , epochs E

- 1: Initialise ContrastiveHead g_ψ randomly
 - 2: Set $\phi \leftarrow \log(0.07)$, freeze ϕ
 - 3: **for** epoch $e = 1, \dots, E$ **do**
 - 4: **if** $e = 3$ **then** unfreeze ϕ
 - 5: **end if**
 - 6: **for** each mini-batch $\{(\mathbf{f}_i, \mathbf{t}_i)\}$ **do**
 - 7: Compute $\mathbf{u}_i, \mathbf{v}_i$ via Eq. (2)
 - 8: Compute $\mathcal{L}$ via Eq. (4)
 - 9: Backprop, clip gradients, step AdamW
 - 10: Clamp $\phi \in [-4.6, 2.3]$
 - 11: **end for**
 - 12: Evaluate val loss; save best checkpoint
 - 13: **end for**
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ViT Freezing. We experiment with two settings: (a) **Frozen ViT**: only ContrastiveHead + ϕ are trained ($\sim 265\text{K}$ parameters), and (b) **Unfrozen ViT**: the full backbone is fine-tuned ($\sim 302\text{M}$ parameters). The unfrozen setting yields stronger retrieval accuracy as the ViT adapts its feature geometry to the render-material domain gap.

Image Pre-processing. Both images are resized to 448×448 via bicubic interpolation and normalised with ImageNet statistics ($\mu = (0.485, 0.456, 0.406)$, $\sigma = (0.229, 0.224, 0.225)$). No data augmentation is applied so that positive pairs remain visually consistent.

Inference. At query time, the rendered image is encoded to a 256-dim vector. The gallery of texture embeddings is pre-computed and stored in a FAISS IndexFlatIP [9] index. Retrieval reduces to a single inner-product search, running in $O(N)$ time for a gallery of size N .

6 Experiments

6.1 Evaluation Protocol

We evaluate on the full validation set of 1,956 (render, texture) pairs. The retrieval gallery consists of all 1,956 texture embeddings. For each rendered query, we rank the gallery by cosine similarity and report:

- **R@K**: fraction of queries for which the ground-truth texture appears in the top- K results ($K = 1, 3, 5, 10$).
- **MRR**: mean of $1/\text{rank}$ over all queries.
- **nDCG@10**: normalised discounted cumulative gain at cutoff 10.
- **MAP@10**: mean average precision at cutoff 10.
- **rSum**: the naive aggregate $R@1 + R@5 + R@10$ (in %).
- **RMR-Score**: our proposed unified metric (Eq. 1).

6.2 Main Results

Table 2 reports retrieval performance on the validation gallery.

Key observations. (1) All zero-shot baselines achieve near-zero RMR-Score (≤ 0.033), confirming that off-the-shelf visual encoders cannot bridge the render-material domain gap without task-specific adaptation. (2) Training only the ContrastiveHead while freezing the ViT (“frozen ViT”) already achieves $\text{RMR} = 0.212$ ($R@1 = 11.0\%$), a meaningful improvement over all zero-shot baselines, indicating that even a lightweight projection head can partially bridge the domain gap when trained on paired data. (3) Unfreezing the ViT backbone yields a further $2.5\times$ **boost**: our full model achieves **$\text{RMR} = 0.524$** , a $15.9\times$ **improvement** over the strongest zero-shot baseline (CLIP ViT-L/14, $\text{RMR} = 0.033$). On individual metrics, $R@1$ improves from 1.8% to 34.5% ($19\times$), $\text{nDCG}@10$ from 0.040 to 0.579 ($14\times$), and median rank drops from 473 to 2. (4) The large gap between $R@1$ and $R@5$ ($34.5\% \rightarrow 71.6\%$) suggests that near-miss rank-1 errors are predominantly same-family confusions between visually similar material variants.

6.3 Metric Comparison

Table 2 also illustrates why a unified metric is valuable: rSum for CLIP ViT-L/14 is 13.2, suggesting reasonable coverage, but RMR-Score correctly reveals this as only 0.033 — reflecting the fact that near-all retrieved results are wrong. Conversely, our model’s rSum of 187.5 and RMR of 0.524 consistently indicate strong performance across all three axes.

6.4 Qualitative Analysis

Inspection of retrieval errors reveals two failure modes:

Same-family confusion. Textures within the same material class (e.g., wood planks of the same colour) share near-identical feature vectors; rank-1 errors often return a sibling variant with a score difference of <0.001 . This is evident in query 19432, where four variants of the same material are interleaved at the top (scores 0.902–0.902).

Appearance–geometry entanglement. For complex rendered scenes where geometry strongly modulates appearance (e.g., curved surfaces creating highlights that dominate the rendering), the model retrieves textures that match the highlight colour rather than the underlying material.

6.5 Ablation: Temperature Warmup

Freezing the temperature τ for the first two epochs prevents it from collapsing to near-zero early in training (which would cause gradient explosion through the logits). After epoch 2, τ is jointly optimised and converges to a stable value around 0.07–0.15 depending on the effective batch size.

7 Conclusion

We have presented ReTR, a contrastive fine-tuning approach that adapts the visual backbone of InternVL2-1B to the render-to-material retrieval task, together with a unified evaluation metric (RMR-Score) that combines precision, coverage, and ranking quality into a single comparable scalar. Our method achieves $\text{RMR-Score} = 0.524$ on a gallery of 1,956 material textures, outperforming all zero-shot baselines by over an order of magnitude. This strong result demonstrates that VLM visual backbones can be efficiently repurposed for domain-specific cross-modal retrieval, and that a unified metric is essential for holistic evaluation.

Future work includes: (i) extending the training set with augmented renders (varied lighting, geometry, viewpoints); (ii) scaling to InternVL2-8B for improved representation capacity; (iii) integrating the retrieval model into an agentic pipeline for end-to-end material recommendation.

References

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Table 2: Retrieval results on the validation set (gallery size = 1,956). $R@K$ in %; MRR, nDCG, MAP $\in [0, 1]$; MedR = median rank ($\downarrow$); rSum = $R@1+R@5+R@10$ ($\uparrow$); RMR = proposed unified metric ($\uparrow$). Best results in **bold**.

Method	Standard metrics					Aggregate metrics			RMR
	R@1	R@5	R@10	MRR	MedR	nDCG@10	MAP@10	rSum	
Random	0.1	0.3	0.5	.004	978	.004	.003	0.8	.002
CLIP ViT-B/32 [1]	1.2	3.1	4.7	.026	500	.026	.020	8.9	.023
CLIP ViT-L/14 [1]	1.8	4.5	7.0	.037	473	.040	.031	13.2	.033
InternVL2-1B (zero-shot)	0.3	1.1	1.5	.011	722	.008	.006	2.9	.008
Ours (frozen ViT)	11.0	31.0	43.3	.216	14	.253	.198	85.3	.212
Ours (unfrozen ViT)	34.5	71.6	81.4	.511	2	.579	.503	187.5	.524

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